The Development and Application of Optogenetics

Paper: Fenno, Yizhar and Deisseroth (2011). Review.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What optical stimulation can control, how targeting is achieved, and which actuator dynamics constrain a neural control model.

How does it answer the general questions?

Actuation. Reviews excitatory channelrhodopsins, inhibitory pumps and light-sensitive signalling tools. Genetic and projection targeting allow selective manipulation; opsin kinetics and spectral sensitivity vary substantially across tools (PDF pp. 2–11).

Usable insight. Light input is filtered by expression, delivery and opsin physiology before affecting spikes or circuit activity. High-frequency stimulation fidelity, adaptation and recovery therefore need validation for the selected construct. Electrical or optical measurements are needed to verify the effect; the actuator is not itself a sensor (PDF pp. 7–16).

Model relevance. Helps choose actuator location, sign and bandwidth and motivates adding opsin states when a static gain is inadequate. It does not establish the notebook's monotone memoryless nonlinearity assumptions, numerical delayed coupling coefficients or an LFP observation map. Discussed behavioural interventions are application-specific; this review does not identify beta suppression as a general therapeutic objective.

What papers does it cite that could provide more information?

- Boyden et al. (2005), *Millisecond-timescale, genetically targeted optical control of neural activity*: optical excitation (references, PDF p. 19).
- Gradinaru et al. (2009), *Optical deconstruction of parkinsonian neural circuitry*: pathway-level intervention (PDF p. 20).